

Week 1 Module - Programming Practice (3rd Year)

1) STAR Patterns.

1. Square

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
import java.util.*;

public class Patterns {
    public static void main (String[] args) {
        int n = 5;
        System.out.println ("Square");
        for (int i = 0; i < n; i++)
            for (int j = 0; j < n; j++)
                System.out.print ("* ");
            System.out.println ();
    }
}
```

2) Hollow Square

```
(1,1) * (1,2) (1,3) (1,4) (1,5)
(2,1) * (2,5)
(3,1) * (3,5)
(4,1) * (4,5)
(5,1) * * * * * (5,5)
```

```
int n = 5;
System.out.println ("Hollow Square");
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        if (i == 0 || i == n - 1 || j == 0 || j == n - 1)
            System.out.print ("* ");
        else
            System.out.print (" ");
    }
    System.out.println ();
}
```

Triangle

```

      * (1,1)
    * * (2,2)
  * * * (3,3)
* * * *
  
```

```

int n = 5;
System.out.println("Triangle");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print("*");
    }
    System.out.println();
}
  
```

Inverted Triangle

```

* * * * *
 * * * *
  * * *
   * *
    *
  
```

```

int n = 5;
System.out.println("Inverted Triangle");
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print("*");
    }
    System.out.println();
}
  
```

Pyramid

```

      *
     * *
    * * *
   * * * *
  * * * * *
  
```

```

int n = 5;
System.out.println("Pyramid");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i + 1; j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= 2 * i - 1; j++) {
        System.out.print("*");
    }
    System.out.println();
}
  
```

6. Diamond

Pyramid + Inverted Triangle

```

      *
     **
    ***
   ****
  *****
 *****
  *****
   ****
    ***
     **
      *
  
```

```

int n = 5;
System.out.println("Diamond");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= 2 * i - 1; j++)
        System.out.print("*");
    System.out.println();
}
  
```

```

}
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++)
        System.out.print(" ");
    for (int j = 1; j <= 2 * i - 1; j++)
        System.out.print("*");
    System.out.println();
}
  
```

7. Half Diamond

Triangle + Lower Triangle

```

      *
     **
    ***
   ****
  *****
 *****
  *****
   ****
    ***
     **
      *
  
```

```

int n = 5;
System.out.println("Half Diamond");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++)
        System.out.print("*");
    System.out.println();
}
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= i; j++)
        System.out.print("*");
    System.out.println();
}
  
```

Number Patterns

① Square

```

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
    
```

```

int n = 5;
System.out.println("Number Square");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        System.out.print("1 ");
    }
    System.out.println();
}
    
```

Incrementing Numbers

```

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
    
```

```

int n = 5;
System.out.println("Increment Numbers");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        System.out.print(j + " ");
    }
    System.out.println();
}
    
```

Right Triangle Numbers

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
    
```

```

int n = 5;
System.out.println("Right ---");
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print(j + " ");
    }
    System.out.println();
}
    
```

Inverted Diamond Numbers

```

3 3 3 3 3
4 4 4 4
5 5 5
6 6
7
    
```

```

int n = 5;
System.out.println("Inverted ---");
int num = 3;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print(num + " ");
    }
    System.out.println();
    num++;
}
    
```

Sandwich Pattern

94

```

3 3 3 3 3
 4 4 4
   5
  4 4 4
3 3 3 3 3
    
```

```

int n = 5;
System.out.println("Sandwich Pattern");
for (int i = 3; i <= 5; i++) {
    for (int j = 1; j <= i; j++)
        System.out.print(i + " ");
    System.out.println();
}
for (int i = 4; i >= 3; i--) {
    for (int j = 1; j <= i; j++)
        System.out.print(i + " ");
    System.out.println();
}
    
```

Increment Diamond Pattern

```

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
    
```

```

int n = 5;
System.out.println("Increment Diam-");
int star = 3;
for (int i = 3; i <= 7; i++) {
    for (int j = 3; j <= i; j++)
        System.out.print(i + " ");
    System.out.println();
}
for (int i = 6; i >= 3; i--) {
    for (int j = 3; j <= i; j++)
        System.out.print(i + " ");
    System.out.println();
}
    
```

Patterns.java



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Run

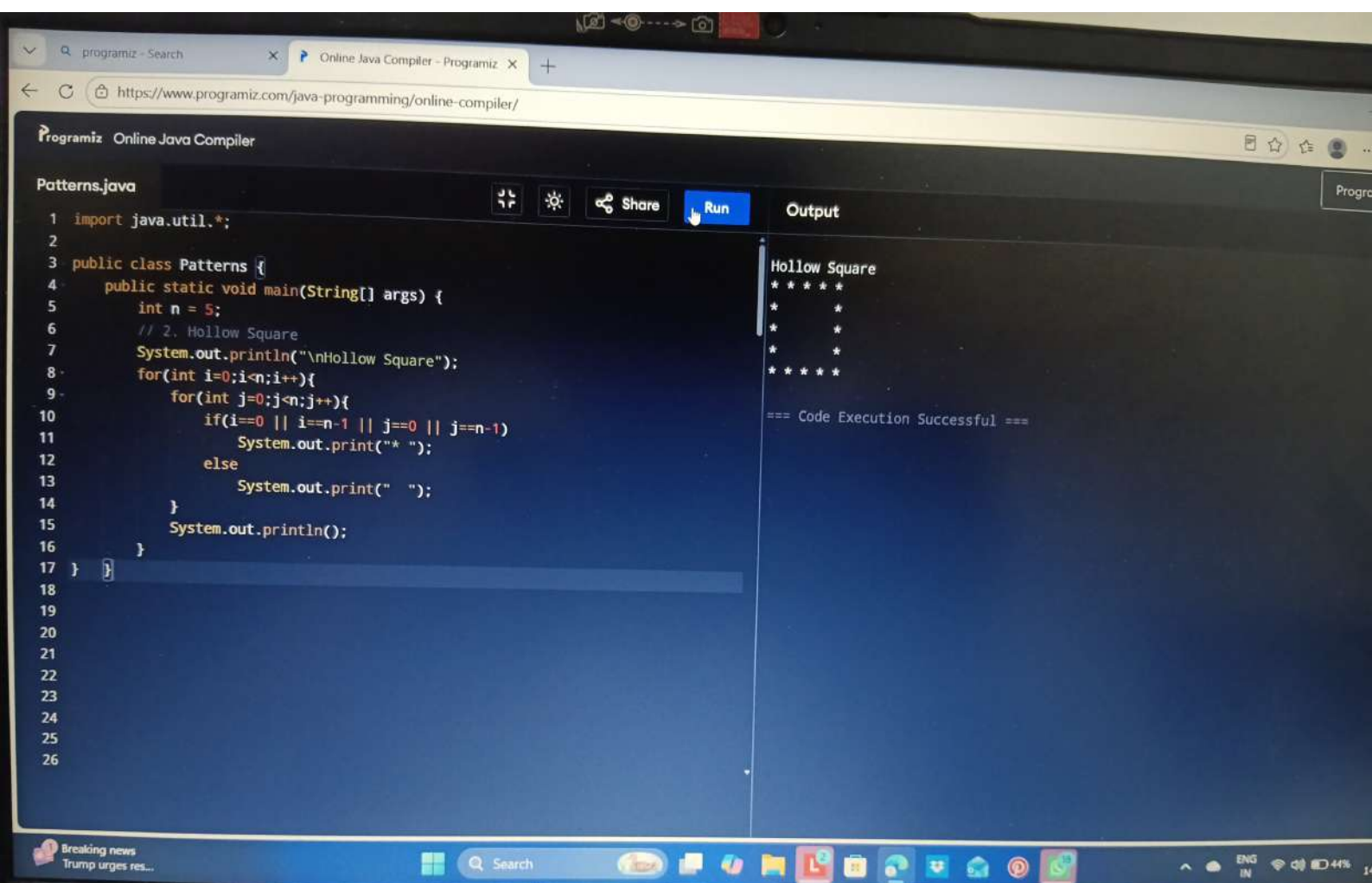
Output

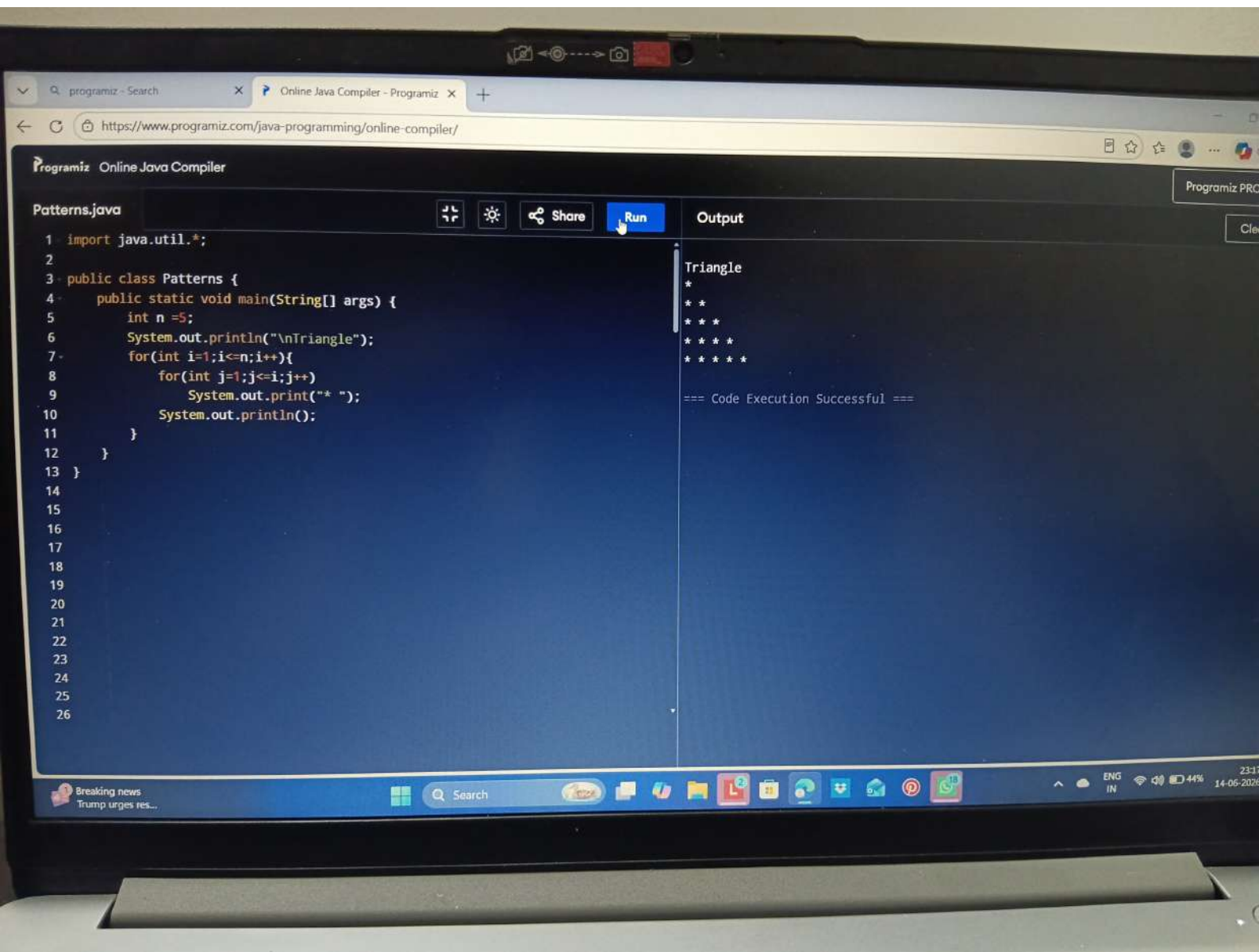
```
1 import java.util.*;
2
3 public class Patterns {
4     public static void main(String[] args) {
5         int n = 5;
6
7         // 1. Square
8         System.out.println("Square");
9         for(int i=0;i<n;i++){
10             for(int j=0;j<n;j++){
11                 System.out.print("* ");
12                 System.out.println();
13             }
14         }
15
16
17
18
19
20
21
22
23
24
25
26
```

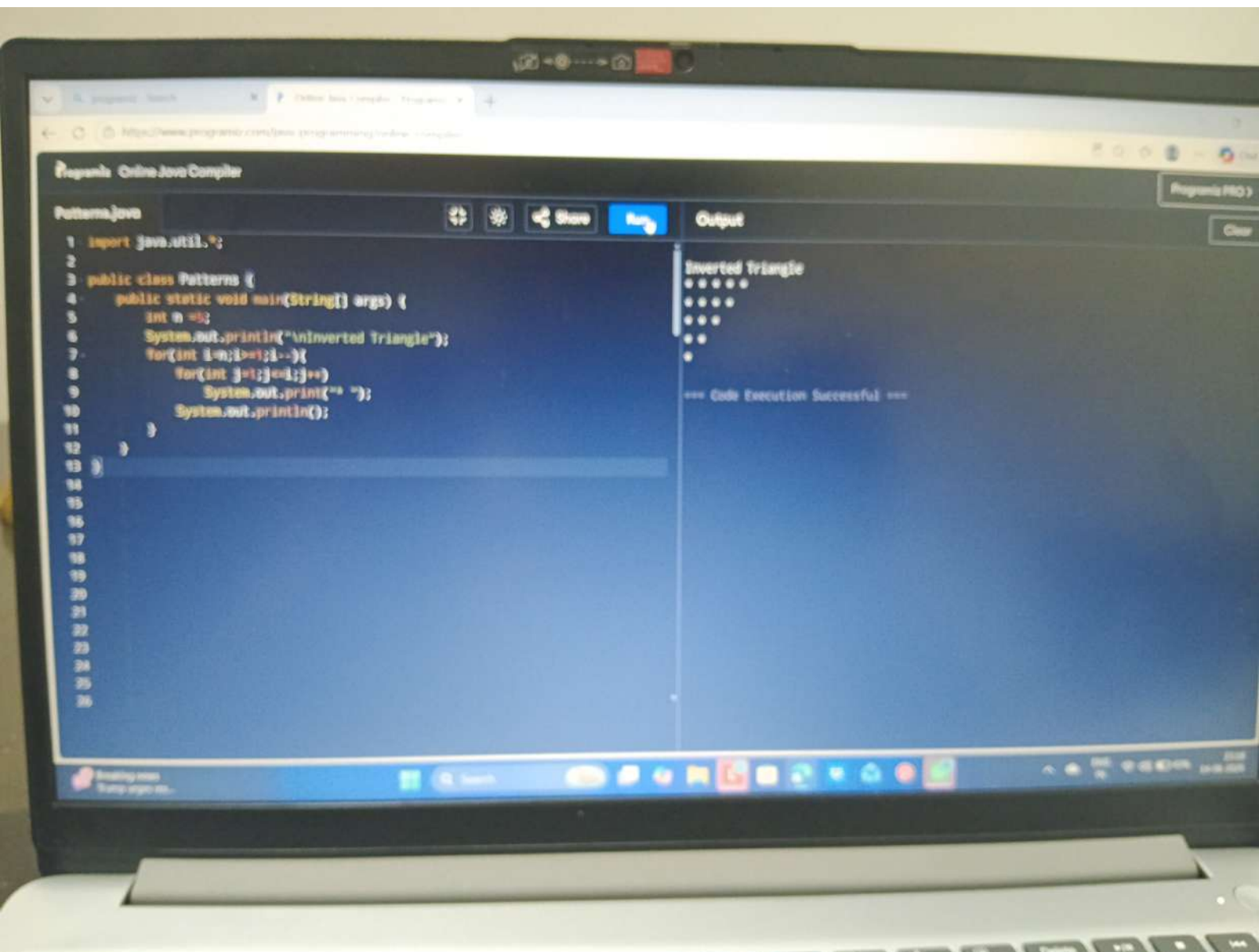
Square

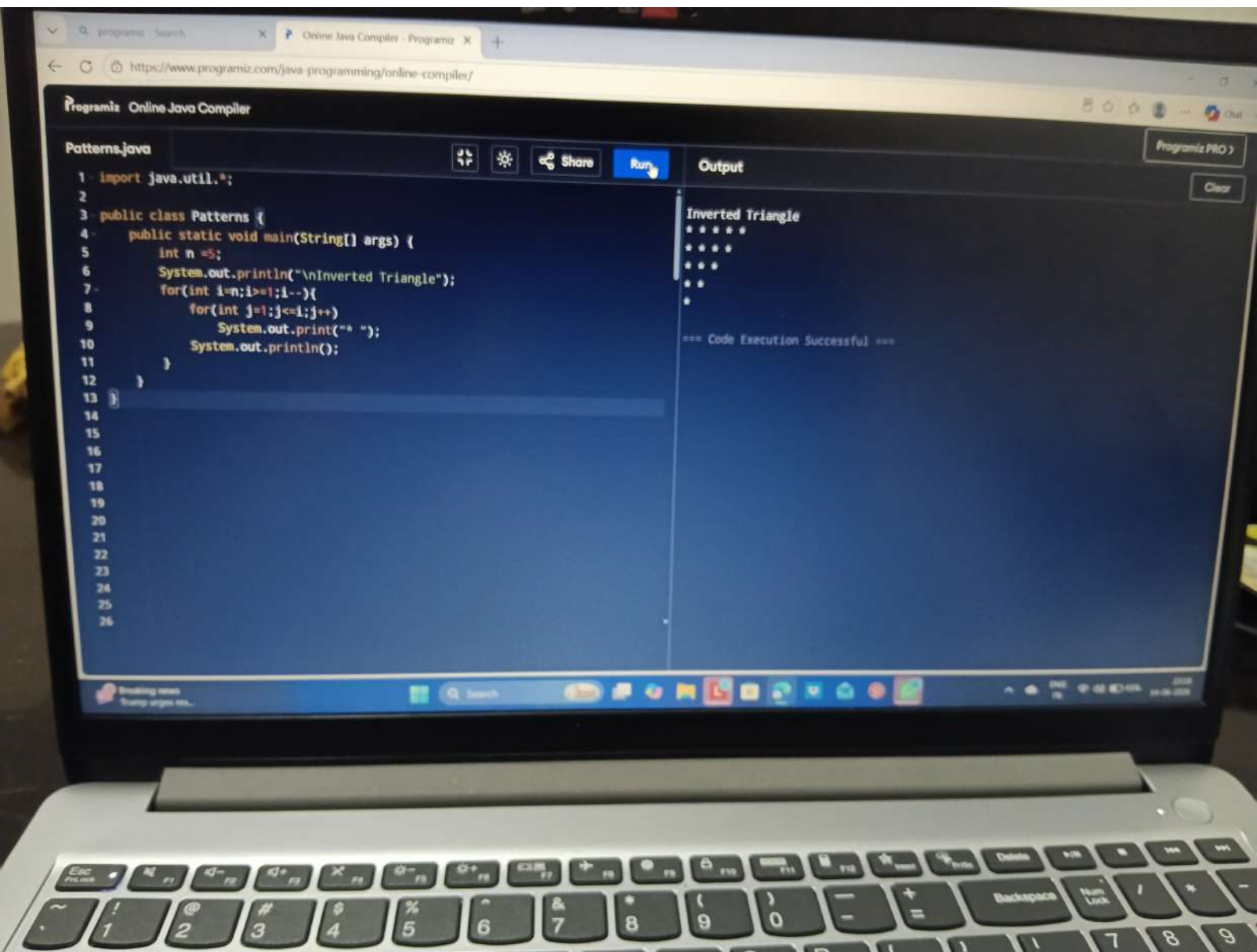
```
*****
*****
*****
*****
*****
```

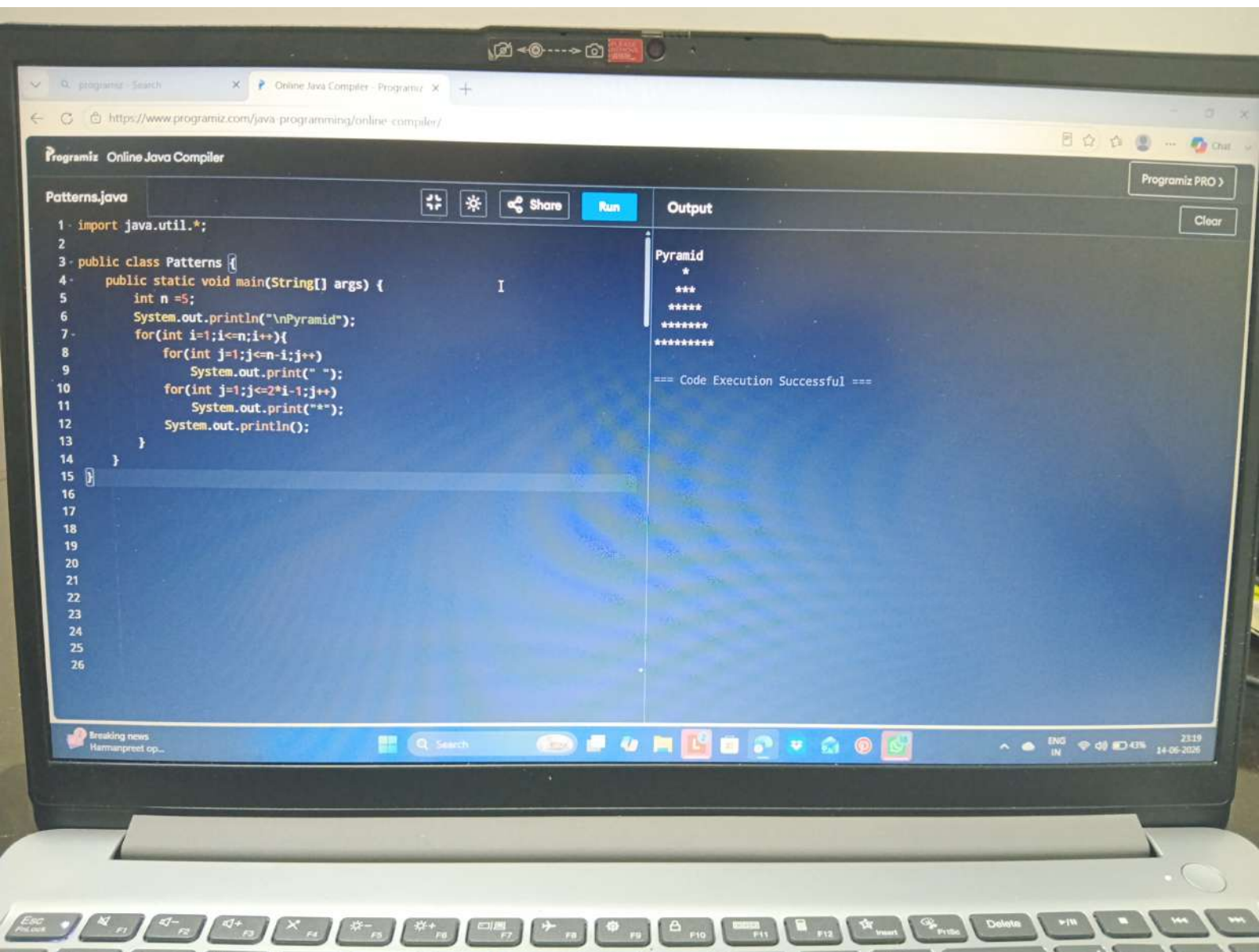
=== Code Execution Successful ===

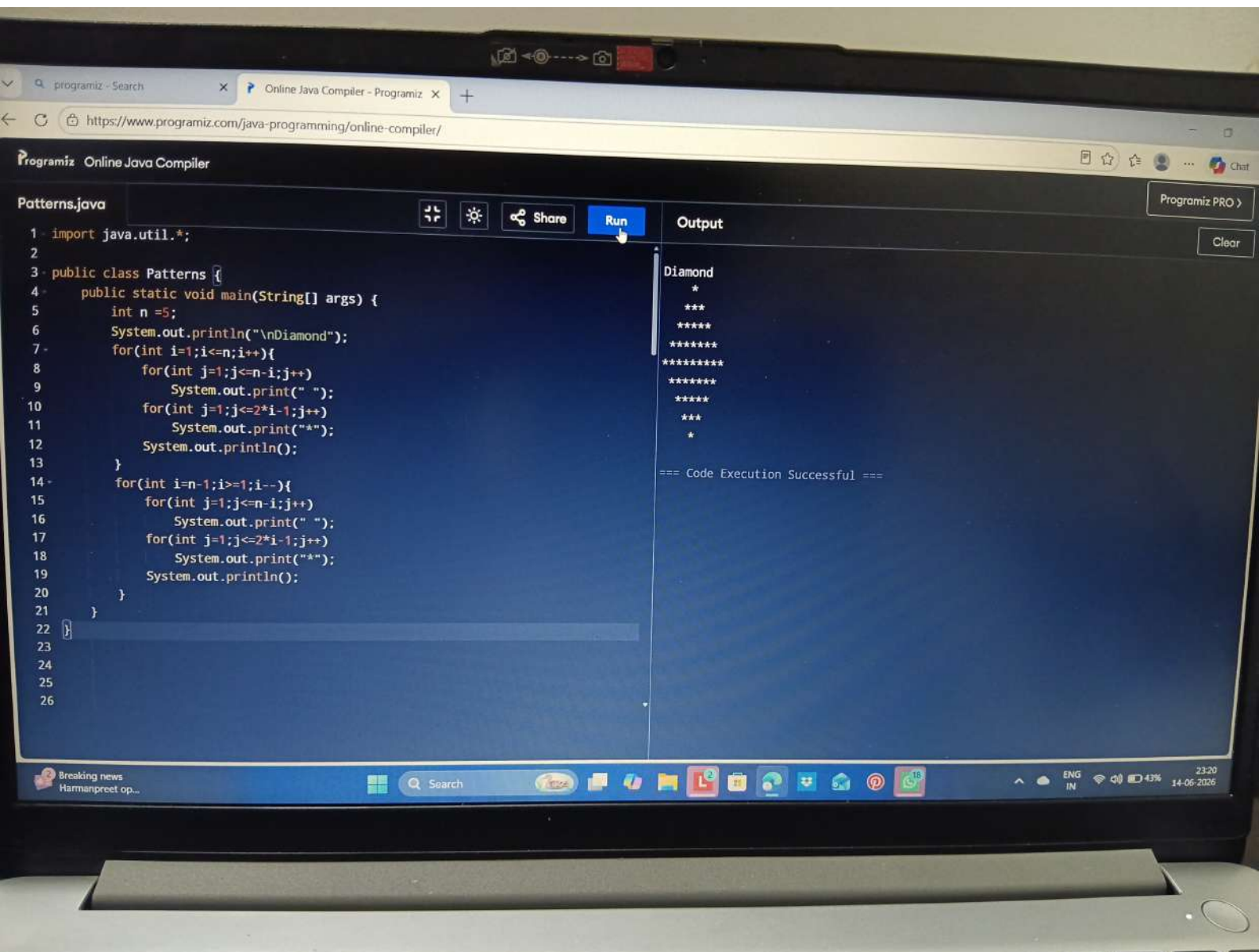


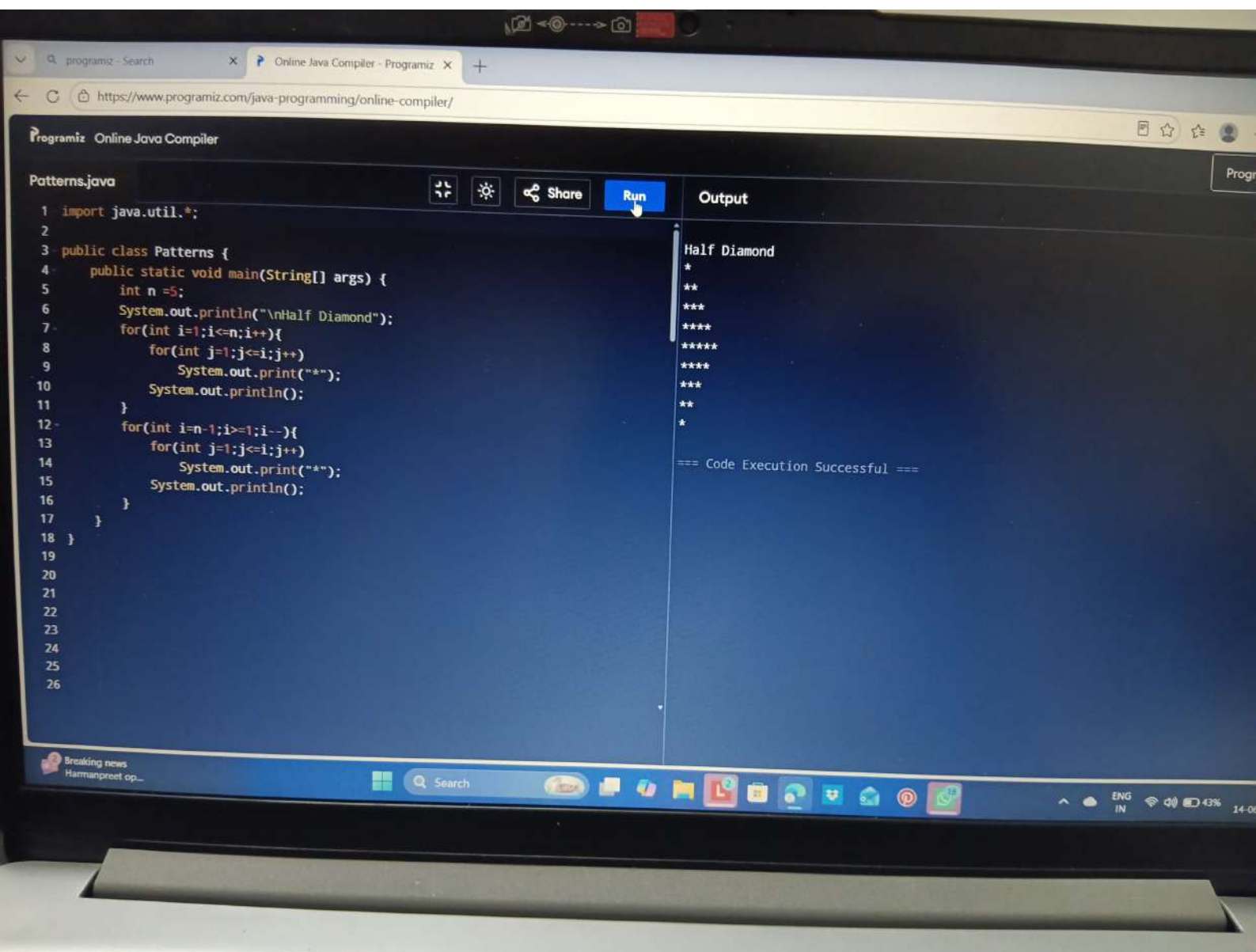












Patterns.java



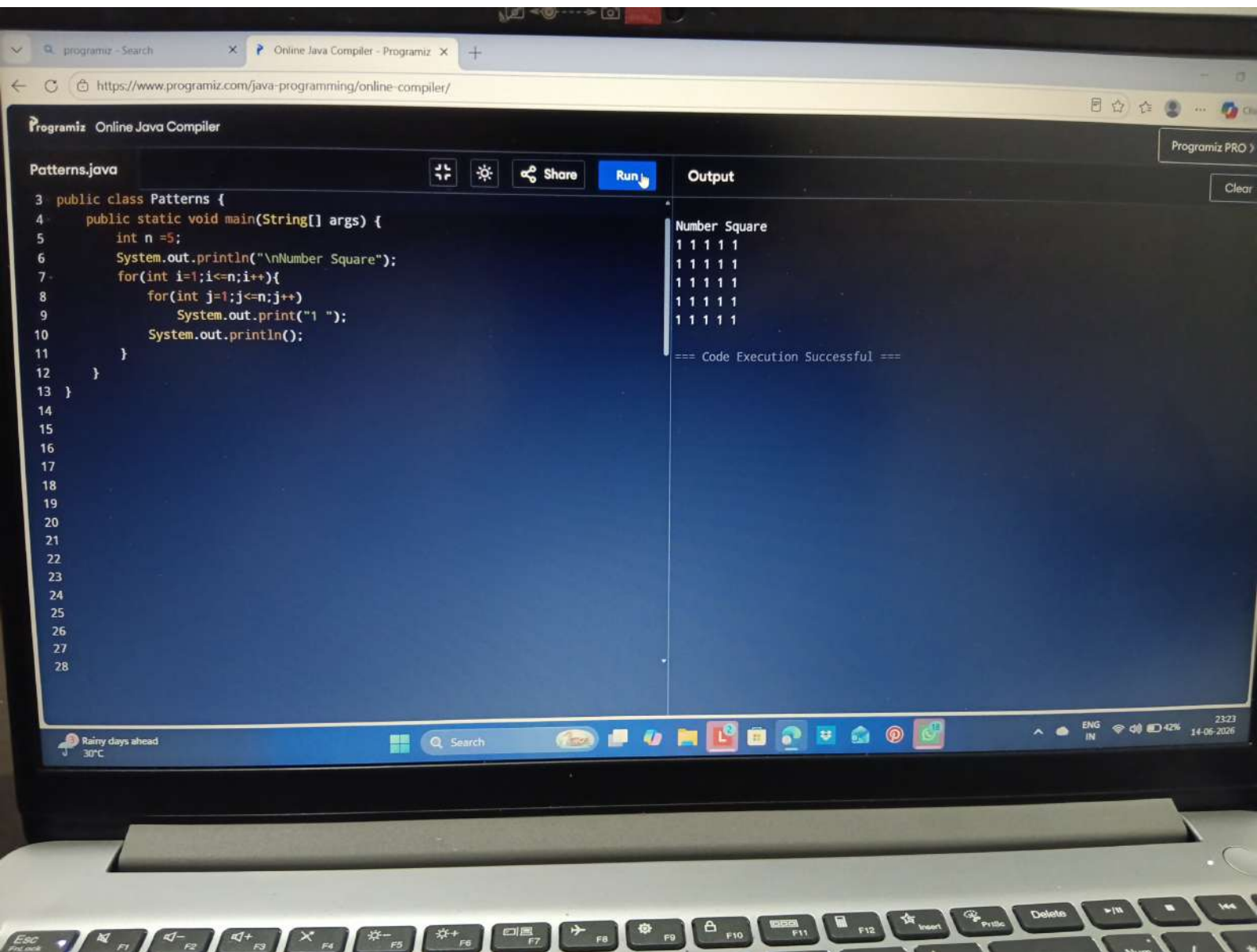
Output

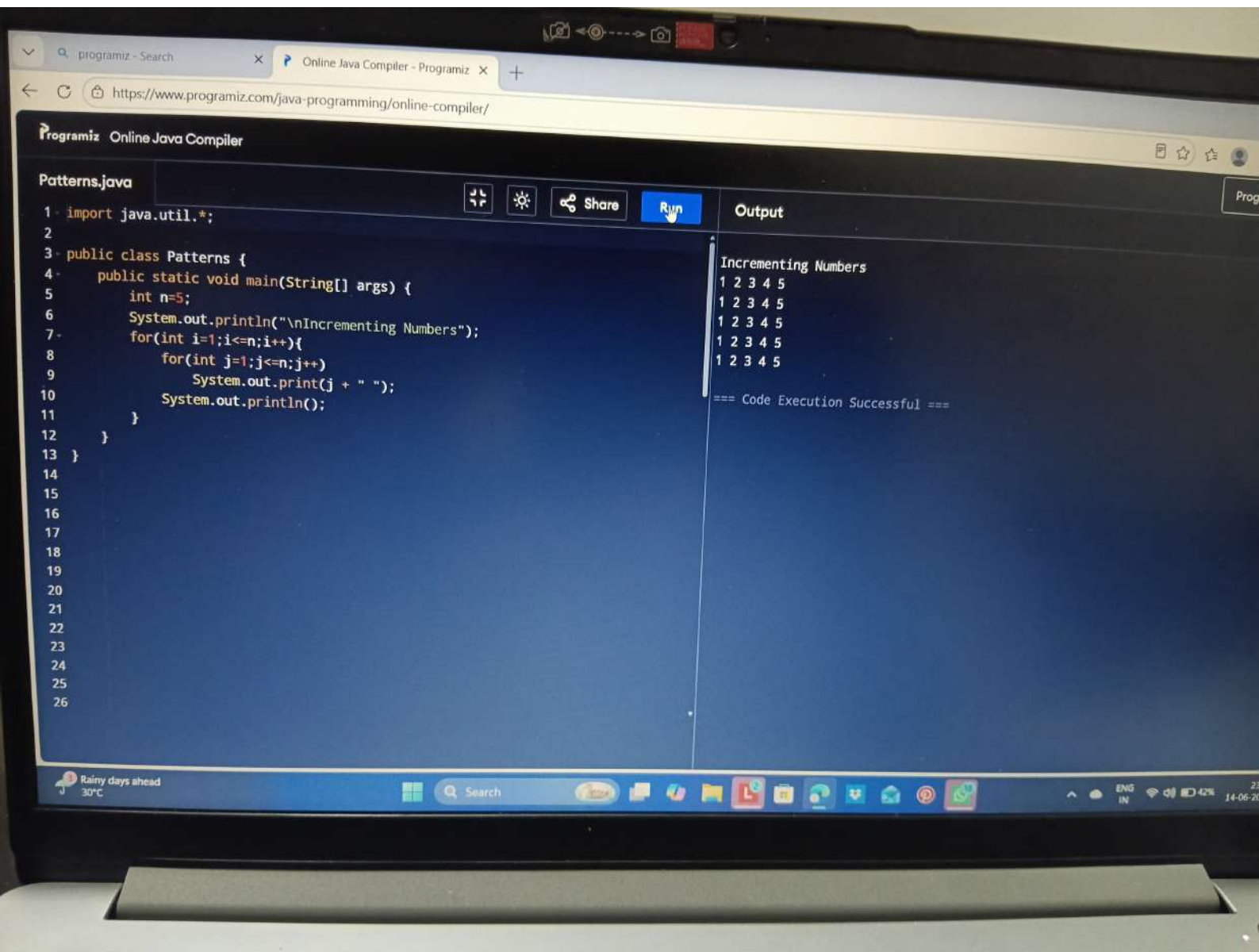
Clear

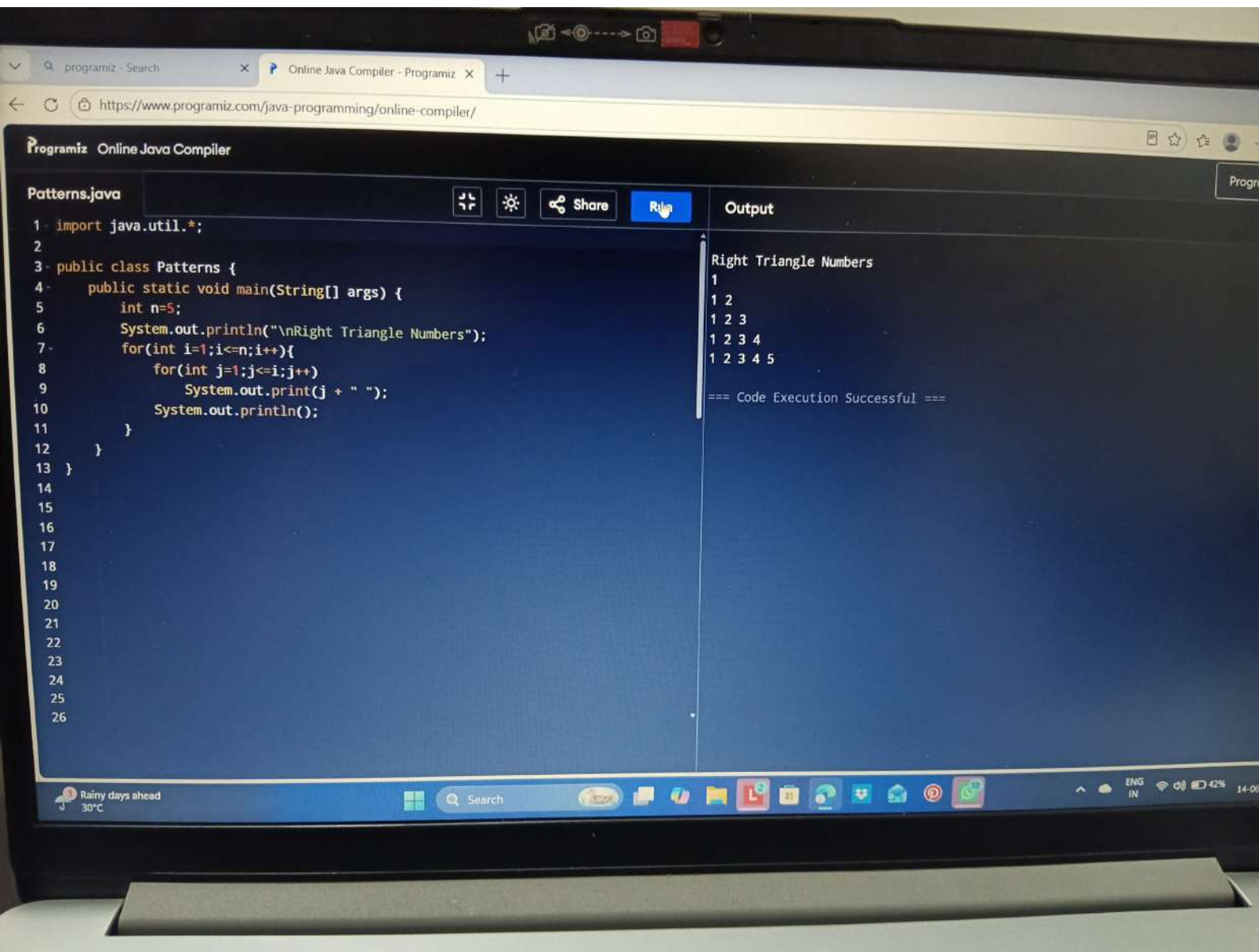
```
1 import java.util.*;
2
3 public class Patterns {
4     public static void main(String[] args) {
5         int n = 5;
6         System.out.println("\nHalf Diamond Inverted");
7         for(int i=n;i>=1;i--){
8             for(int j=1;j<=i;j++){
9                 System.out.print("*");
10                System.out.println();
11            }
12        }
13    }
14
15
16
17
18
19
20
21
22
23
24
25
26
```

```
Half Diamond Inverted
*****
****
***
**
*
```

=== Code Execution Successful ===







Patterns.java



Output

Clear

```
1- import java.util.*;
2
3- public class Patterns {
4-     public static void main(String[] args) {
5-         int n=5;
6-         System.out.println("\nInverted Diamond Numbers");
7-         int num = 3;
8-         for(int i=n;i>=1;i--){
9-             for(int j=1;j<=i;j++){
10-                 System.out.print(num + " ");
11-                 System.out.println();
12-                 num++;
13-             }
14-         }
15-     }
16- }
17
18
19
20
21
22
23
24
25
26
```

Inverted Diamond Numbers

3 3 3 3 3

4 4 4 4

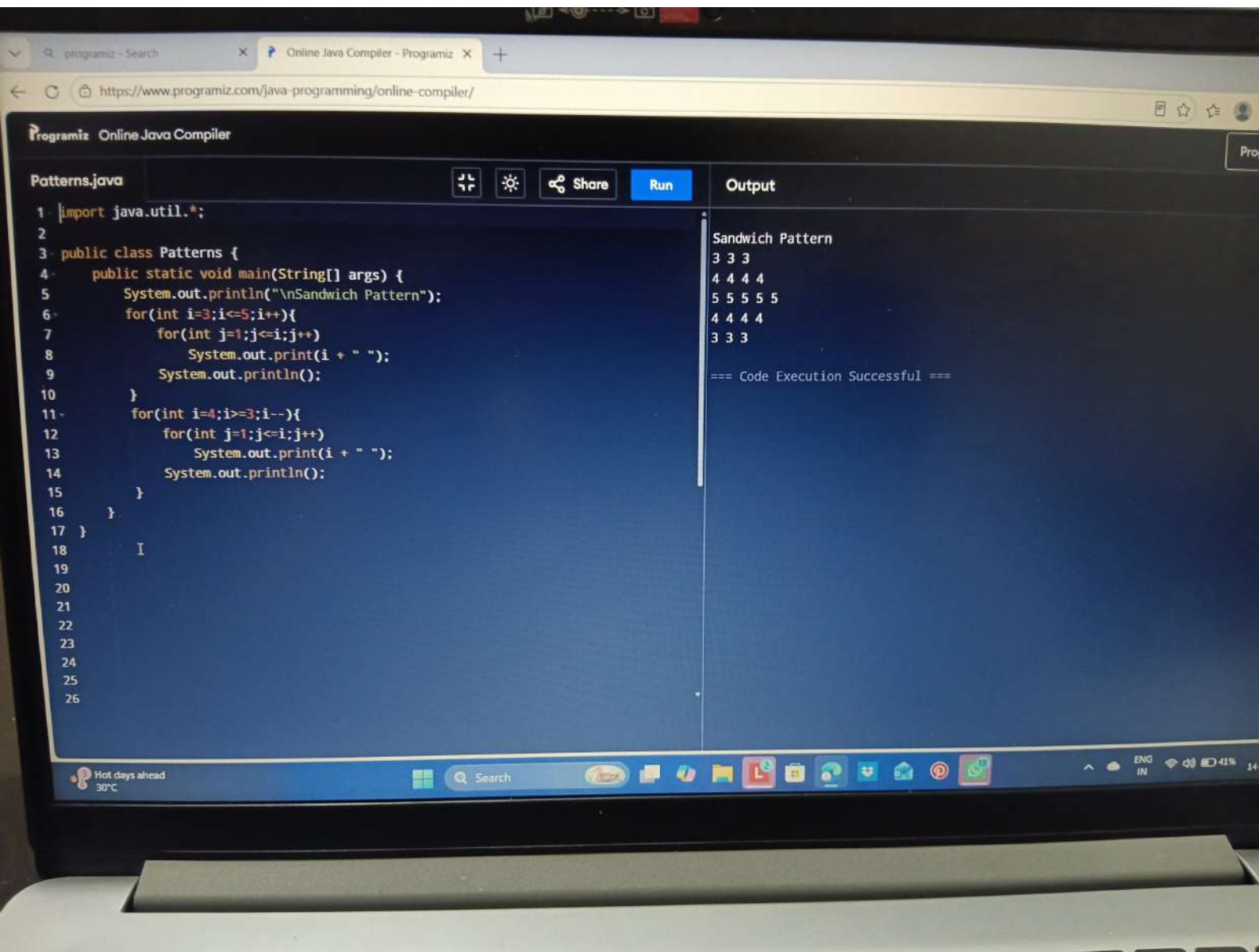
5 5 5

6 6

7

=== Code Execution Successful ===





Patterns.java

```
1 import java.util.*;
2
3 public class Patterns {
4     public static void main(String[] args) {
5         System.out.println("\nIncrementing Diamond Pattern");
6         int start = 3;
7         for(int i=3;i<=7;i++){
8             for(int j=3;j<=i;j++)
9                 System.out.print(i + " ");
10            System.out.println();
11        }
12        for(int i=6;i>=3;i--){
13            for(int j=3;j<=i;j++)
14                System.out.print(i + " ");
15            System.out.println();
16        }
17    }
18 }
```

Output

Incrementing Diamond Pattern
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

=== Code Execution Successful ===